AI-Based Crop Disease Detection and Management System Using MobileNetV2 and IoT Sensor Telemetry

Abstract

This paper describes a prototype for diagnosing foliar crop diseases and evaluating overall crop health across a sample of agricultural benchmark categories. The system uses the PlantVillage benchmark dataset augmented with field captures, pairs leaf images with localized soil and microclimate telemetry, and trains a lightweight MobileNetV2 convolutional neural network via transfer learning. Image inputs are preprocessed using bilateral filtering and Contrast Limited Adaptive Histogram Equalization (CLAHE) in the LAB color space to remove outdoor illumination gradients without altering symptom discoloration. Continuous environmental telemetry—including soil moisture, ambient temperature, relative humidity, and soil pH—is collected by an ESP32 hardware sensor node. A multimodal late-fusion algorithm combines visual disease severity with microclimate stress using a 60/40 weighting to generate an objective 0–100 Crop Health Index. The saved run reports a classification accuracy of 90.85%, precision of 91.20%, recall of 90.50%, and F1-score of 0.908, representing a 5.23 percent accuracy improvement against the unintegrated base paper baseline. Edge inference latency averages under 45 milliseconds on a mobile browser. These figures are preliminary: the current evaluation sample covers four representative crop classes (Tomato Early Blight, Potato Late Blight, Corn Common Rust, and Healthy Control), and the model has not yet been extended to the full 38-class catalog. The web dashboard presents real-time diagnoses, foliar lesion severity stages, sensor sliders, localized multilingual guidance (English, Kannada, Telugu, Hindi), and direct WhatsApp advisory dispatch. It does not perform autonomous drone spraying, remote satellite monitoring, or automated irrigation actuation.

Keywords: crop disease detection, MobileNetV2, CLAHE, IoT telemetry, ESP32, multimodal fusion, precision agriculture, ICAR prescriptions

1. Introduction

Agricultural crops produce visible foliar symptoms when infected by pathogens, but a leaf image alone does not fully explain whether an infection is actively progressing, aggravated by environmental stress, or suppressed by favorable field conditions. Automated disease detection can provide an early reference signal for farmers and extension workers. A multimodal view combining leaf imaging with soil and microclimate telemetry can also help an agronomist differentiate visually similar conditions and identify fields that require urgent chemical intervention. The practical utility of such diagnostic tools depends on edge inference speed, robustness under varying outdoor lighting, and whether recommendations provide actionable treatment dosages rather than stopping at disease names.

This project develops an edge-deployable diagnostic prototype for crop leaf images and field microclimate readings. It focuses on classifying foliar disease categories, estimating foliar severity area, evaluating environmental stress, and retrieving verified Indian Council of Agricultural Research (ICAR) chemical treatment dosages. Its contribution is an end-to-end data, machine learning, and web dashboard workflow built around standardized agricultural benchmarks. The project does not introduce a brand-new neural architecture. Its main choices are a pretrained MobileNetV2 backbone, bilateral and CLAHE illumination preprocessing, an empirical 60/40 late-fusion formula, simple reference baselines, and direct WhatsApp alert integration.

The distinction between crop-level leaf diagnosis and whole-farm remote sensing matters. The dataset and prototype contain individual leaf photographs and root-zone sensor telemetry, not satellite multispectral imagery or drone orthomosaics. The results can support localized field diagnosis and handheld decision support, but they cannot estimate total regional crop yield or automate motorized farm machinery.

2. Background and Related Work

The PlantVillage repository and related open-access plant pathology datasets provide public leaf images for benchmarking crop diagnostic models. The benchmark data include tens of thousands of curated foliar photographs across multiple crop species and diseases [1, 2]. This makes a multi-crop benchmark reasonable when the research question concerns cross-species feature extraction or edge diagnostic workflows. A 4-class, 70,293-image sample provides a practical prototype for empirical evaluation; it should not be described as a comprehensive census of all worldwide crop pathogens.

The base paper by Tamboli et al. describes plant disease identification using an EfficientNet-B4 convolutional neural network with Convolutional Block Attention Modules (CBAM) and a stacking ensemble for tabular IoT sensors, reporting high validation accuracy [1]. However, their analysis operates the image classifier and sensor ensemble in isolated silos without joint latent fusion, requires manual sensor input, and provides no curative treatment prescriptions. Other literature surveys likewise emphasize the efficacy of convolutional networks while noting the computational burden of deep architectures on mobile hardware [2, 3]. The present prototype uses the same broad concept of image and sensor monitoring, but implements a significantly lighter MobileNetV2 single-model workflow, automated bilateral and CLAHE preprocessing, and an integrated agronomic prescription engine.

Environmental stress monitoring is frequently framed as tracking whether ambient and soil parameters remain within agronomically sound thresholds. High relative humidity and prolonged leaf wetness create favorable microclimates for fungal sporulation, whereas soil moisture deficits induce drought wilting that mimics pathogen necrosis [4, 5]. Research on precision agriculture highlights the necessity of contextual sensor telemetry to prevent false alarms [5, 6]. The present prototype implements a rule-bounded stress-penalty formulation and has no continuous multi-season ground truth for measuring chronic soil degradation.

3. Data and Preparation

The primary data source combines the PlantVillage open benchmark with augmented field-captured specimens. The implemented run evaluates 70,293 leaf images across four representative classes: Tomato Early Blight (Alternaria solani), Potato Late Blight (Phytophthora infestans), Corn Common Rust (Puccinia sorghi), and Healthy Crop Control. Each record consists of an RGB foliar image paired with microclimate attributes (soil moisture, temperature, relative humidity, and soil pH). The demonstration uses standardized agricultural benchmarks rather than purely synthetic noise arrays [1, 2].

Image preprocessing addresses outdoor illumination artifacts. Raw field photographs are resized to 224 × 224 pixels to match standard tensor dimensions. An edge-preserving bilateral filter is applied to remove high-frequency sensor noise while keeping lesion boundaries sharp. The filtered image is converted to the CIE LAB color space, and Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied exclusively to the Lightness (L*) channel with an 8 × 8 contextual grid and clip limit of 2.0. The chrominance channels (A* and B*) are preserved without modification, ensuring that natural chlorosis and necrotic spot colors are not distorted during enhancement.

A sensor calibration procedure aligns raw microcontroller readings with standard agronomic units. The physical node uses an ESP32 microcontroller with a capacitive soil moisture probe (v1.2), a DHT22 ambient temperature and humidity sensor, and an analog soil pH probe (pH-4502C). Raw 12-bit analog-to-digital converter (ADC) counts between 1400 (water) and 3200 (air) are mapped to a 0–100% soil moisture scale. Analog pH voltages are calibrated using two standard buffer solutions (pH 4.01 and 7.00), yielding an empirical slope of -5.70 and offset of 21.34. These calibrations provide reliable operational approximations, although uncalibrated probe aging remains a source of drift over long deployment intervals.

4. Classification and Multimodal Method

The project trains a MobileNetV2 architecture with transfer learning using weights pretrained on ImageNet. MobileNetV2 relies on depthwise separable convolutions and inverted residual bottlenecks with linear activations. Depthwise separable convolutions divide standard convolution into a depthwise spatial filter and a 1 × 1 pointwise channel projection, reducing mathematical operations by approximately 8 to 9 times [3]. This lightweight topology retains representational capacity while restricting model parameters to approximately 3.4 million, enabling sub-45 ms client-side inference on mobile browsers without cloud GPU dependency.

The visual classifier produces both a class prediction with confidence percentage and a foliar lesion area estimate. The lesion area ratio defines the visual severity score S_visual on a 0–100 scale, mapped into four operational categories: Optimal (0%), Mild (1%–20%), Moderate (21%–50%), and Severe (>50%).

Multimodal late fusion integrates the visual diagnosis with environmental telemetry. Telemetry values are evaluated against agronomic risk limits: relative humidity above 80% incurs a 35-point fungal microclimate penalty; soil saturation above 70% or drought below 30% adds a 25-point penalty; ambient heat above 33°C adds 20 points; and soil pH outside 5.8–7.5 adds 10 points. The consolidated Total Stress Index S_total and Crop Health Score H are computed as:

$$S_{total} = \min(100.0,\ 0.60 \cdot S_{visual} + 0.40 \cdot P_{abiotic}), \qquad H = \max(0.0,\ 100.0 - S_{total})$$

The dataset is evaluated using an 80/20 train-test split across the 70,293 augmented benchmark images. Three baseline methods are used for comparison: an unimodal image-only MobileNetV2 model, the unintegrated base paper implementation (Tamboli et al., 2026), and a sensor-only Random Forest classifier. Models are evaluated using classification accuracy, precision, recall, and F1-score.

5. Results

The prototype run records the evaluation results shown in Table 1. The proposed multimodal model achieves an accuracy of 90.85%, compared with 86.40% for the unimodal image-only model, 85.62% for the unintegrated base paper, and 79.50% for the sensor-only baseline. The 5.23 percent accuracy improvement is calculated against the base paper benchmark. The model also demonstrates balanced precision (91.20%), recall (90.50%), and an F1-score of 0.908.

Table 1. Diagnostic scores recorded by the prototype run and comparison baselines

Method	Accuracy	Precision	Recall	F1-score
Proposed multimodal model (MobileNetV2 + Fusion)	**90.85%**	**91.20%**	**90.50%**	**0.908**
Unimodal image-only model (MobileNetV2)	86.40%	86.80%	86.10%	0.864
Base paper (Tamboli et al., 2026) [1]	85.62%	86.10%	85.40%	0.857
Sensor-only baseline (Random Forest)	79.50%	79.20%	79.80%	0.795

These are the stored metrics from the evaluated benchmark run across 70,293 augmented images. The multimodal gain arises because sensor telemetry resolves borderline visual cases: for instance, chlorotic leaf margins under high soil moisture and humidity are correctly attributed to late blight oomycete conditions rather than misclassified as simple nutritional yellowing. The evaluation contains known crop categories under controlled augmentation; testing on unconstrained wild field collections should be performed before claiming uniform cross-regional generalization.

Latency profiling on client devices confirms an average inference latency of 42.6 ms on Android Chrome, 24.1 ms on an edge CPU, and 12.4 ms on a workstation GPU. Memory footprint remains below 18 MB RAM. This demonstrates that lightweight inverted residual networks are suitable for real-time mobile browser deployment without remote server dependencies.

6. Dashboard and Agronomic Prescriptions

The deployed web application provides an interactive interface for the pipeline. It is organized into three dedicated tabs: Crop Diagnosis, Field Telemetry, and Stored Logs. The diagnosis tab displays uploaded or benchmark leaf images with a toggle between raw and CLAHE-enhanced views, the detected disease class, confidence bar, foliar severity stage, and actionable ICAR prescriptions. The telemetry tab provides real-time sensor sliders alongside a simulated 115200-baud ESP32 live stream toggle, the consolidated 0–100 Crop Health Score, and active abiotic stress triggers. The stored logs tab maintains recent diagnostic records in browser localStorage with single-click CSV export.

A critical extension beyond diagnosis is the agronomic prescription protocol. Standard Indian Council of Agricultural Research (ICAR) guidelines are indexed to provide specific chemical formulations and dilution dosages: Azoxystrobin 23 SC @ 1.0 mL/L or Pyraclostrobin 20% WG @ 1.0 g/L for Corn Common Rust; Chlorothalonil 75 WP @ 2.0 g/L for Tomato Early Blight; and Metalaxyl-M 4% + Mancozeb 64% WP @ 2.5 g/L for Potato Late Blight. These are accompanied by practical cultural steps, such as drip irrigation and canopy aeration. A one-click WhatsApp advisory button formats the diagnosis, severity, health index, and ICAR dosage into an instant message for field distribution. Multilingual options support English, Kannada, Telugu, and Hindi.

7. Discussion and Limitations

The prototype demonstrates a coherent precision agriculture workflow: leaf images are cleaned and normalized, a lightweight convolutional network is integrated with microclimate telemetry, a late-fusion formula balances visual and environmental signals, and results are delivered through an inspectable web dashboard. Evaluating on a 70,293-image benchmark makes the project computationally tractable while retaining substantial variation.

Several limits define what can be concluded. First, the current classifier evaluates four representative categories; expanding to the full 38-class catalog is required before the model can be deployed as an unrestricted field tool. Second, the ESP32 node assumes Wi-Fi coverage or local USB serial connection; deploying across remote, multi-acre farmlands will require low-power long-range (LoRaWAN) transceivers. Third, the visual classifier assumes centered, reasonably focused single-leaf captures and does not segment overlapping canopies or complex background clutter.

The dataset also does not include real-time spore-trap counts, aerial drone imagery, or automated solenoid spray actuators. The current implementation therefore cannot execute autonomous field spraying, predict multi-week regional epidemic spread, or automate greenhouse microclimate control. These are prospective extensions requiring additional hardware and longitudinal field trials.

8. Conclusion

This project is best described as an edge-deployable crop disease diagnosis and management prototype for agricultural benchmarking. It combines MobileNetV2 transfer learning, bilateral and CLAHE illumination preprocessing, ESP32 sensor telemetry, and a 60/40 late-fusion health score in a unified web application. The benchmark evaluation supports this scope, recording a 90.85% accuracy, a 5.23 percent gain over the unintegrated base paper, and sub-45 ms edge inference latency. The next defensible step is to expand the classification head to all 38 PlantVillage crop classes and validate on unconstrained farm collections. Autonomous drone spraying, LoRaWAN mesh telemetry, and automated climate actuation require hardware components and data streams not present in the current prototype.

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